*DESIGN AND ANALYSIS OF ALGORITHM*

*competitive programming*

# QUESTION 1:

## Finding Duplicates-O(n^2) Time Complexity,O(1) Space Complexity

**Find Duplicate in Array.**

**Given a read only array of n integers between 1 and n, find one number that repeats.**

**Input Format:**

**First Line - Number of elements**

**n Lines - n Elements**

**Output Format:**
**Element x - That is repeated**

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:**

```
#include <stdio.h>


int main() {
   int n;


   scanf("%d", &n);
```

```
    int a[n];

    for (int i = 0; i < n; i++) {
        scanf("%d", &a[i]);
    }

    // Compare every element with the elements after it
    for (int i = 0; i < n; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] == a[j]) {
                printf("%d", a[i]);
                return 0;
            }
        }
    }

    return 0;
}
```

**RESULT:**
Thus the code is verified and the expected output is obtained.

# QUESTION 2:

# Finding Duplicates-O(n) Time Complexity,O(1) Space Complexity

**Find Duplicate in Array.**

**Given a read only array of n integers between 1 and n, find one number that repeats.**

**Input Format:**

**First Line - Number of elements**

**n Lines - n Elements**

**Output Format:**
**Element x - That is repeated**

**For example:**

| Input | Result |
|---|---|
| 5<br>1 1 2 3 4 | 1 |

**Answer:**

```
#include <stdio.h>


int main() {
   int n;
   scanf("%d", &n);


   int a[n];


   for (int i = 0; i < n; i++)
      scanf("%d", &a[i]);


   int slow = a[0];
   int fast = a[0];


   do {
      slow = a[slow];
      fast = a[a[fast]];
   } while (slow != fast);


   slow = a[0];
```

```
    while (slow != fast) {
        slow = a[slow];
        fast = a[fast];
    }

    printf("%d", slow);

    return 0;
}
```

**RESULT:**
Thus the code is verified and the expected output is obtained.

# QUESTION 3:

# Print Intersection of 2 sorted arrays-O(m*n)Time Complexity,O(1) Space Complexity

**Find the intersection of two sorted arrays.**
**OR in other words,**
**Given 2 sorted arrays, find all the elements which occur in both the arrays.**
**Input Format**
- **The first line contains T, the number of test cases. Following T lines contain:**

1. **Line 1 contains N1, followed by N1 integers of the first array**
2. **Line 2 contains N2, followed by N2 integers of the second array**

**Output Format**
**The intersection of the arrays in a single line**
**Example**

**Input:**

**1**

**3 10 17 57**

**6 2 7 10 15 57 246**

**Output:**

**10 57**

**Input:**

**1**

**6 1 2 3 4 5 6**

**2 1 6**

**Output:**

**1 6**

**For example:**

| Input | Result |
| --- | --- |
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:**

```
#include <stdio.h>


int main() {
   int T;
   scanf("%d", &T);


   while (T--) {
      int m, n;
      scanf("%d", &m);


      int a[m];
      for (int i = 0; i < m; i++)
```

```
        scanf("%d", &a[i]);

    scanf("%d", &n);

    int b[n];
    for (int i = 0; i < n; i++)
        scanf("%d", &b[i]);

    for (int i = 0; i < m; i++) {
        for (int j = 0; j < n; j++) {
            if (a[i] == b[j]) {
                printf("%d ", a[i]);
                break;
            }
        }
    }

    printf("\n");
  }

  return 0;
}
```

**RESULT:**
Thus the code is verified and the expected output is obtained.

# QUESTION 4:

# Print Intersection of 2 sorted arrays-

# O(m+n)Time Complexity,O(1) Space Complexity

**Find the intersection of two sorted arrays.**

**OR in other words,**

**Given 2 sorted arrays, find all the elements which occur in both the arrays.**

**Input Format**

**· The first line contains T, the number of test cases. Following T lines contain:**

**1. Line 1 contains N1, followed by N1 integers of the first array**

**2. Line 2 contains N2, followed by N2 integers of the second array**

**Output Format**

**The intersection of the arrays in a single line**

**Example**

**Input:**

**1**

**3 10 17 57**

**6 2 7 10 15 57 246**

**Output:**

**10 57**

**Input:**

**1**

**6 1 2 3 4 5 6**

**2 1 6**

**Output:**

**1 6**

**For example:**

| Input | Result |
|---|---|
| 1<br>3 10 17 57<br>6<br>2 7 10 15 57 246 | 10 57 |

**Answer:**

```
#include <stdio.h>
int main() {
   int T;
   scanf("%d", &T);

   while (T--) {
      int m, n;
      scanf("%d", &m);

      int a[m];
      for (int i = 0; i < m; i++)
         scanf("%d", &a[i]);

      scanf("%d", &n);

      int b[n];
      for (int i = 0; i < n; i++)
         scanf("%d", &b[i]);

      int i = 0, j = 0;

      while (i < m && j < n) {
         if (a[i] == b[j]) {
            printf("%d ", a[i]);
            i++;
            j++;
         }
         else if (a[i] < b[j]) {
            i++;
         }
```

```
        else {
            j++;
        }
      }

      printf("\n");
   }

   return 0;
}
```

**RESULT:**
Thus the code is verified and the expected output is obtained.

# QUESTION 5:

# Pair with Difference-O(n^2)Time Complexity,O(1) Space Complexity

**Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.**

**Input Format:**

**First Line n - Number of elements in an array**

**Next n Lines - N elements in the array**

**k - Non - Negative Integer**

**Output Format:**
**1 - If pair exists**

**0 - If no pair exists**

**Explanation for the given Sample Testcase:**

**YES as 5 - 1 = 4**

**So Return 1.**

**For example:**

| Input | Result |
| --- | --- |
| 3<br>1 3 5<br>4 | 1 |

**Answer:**

```
#include <stdio.h>


int main() {
   int n, k;


   scanf("%d", &n);


   int a[n];


   for (int i = 0; i < n; i++)
      scanf("%d", &a[i]);


   scanf("%d", &k);


   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         if (i != j && a[j] - a[i] == k) {
            printf("1");
            return 0;
         }
      }
   }
```

```
    printf("0");


    return 0;
}
```

**RESULT:**
Thus the code is verified and the expected output is obtained.

# QUESTION 6:

# Pair with Difference -O(n) Time Complexity,O(1) Space Complexity

**Given an array A of sorted integers and another non negative integer k, find if there exists 2 indices i and j such that A[j] - A[i] = k, i != j.**

**Input Format:**

**First Line n - Number of elements in an array**

**Next n Lines - N elements in the array**

**k - Non - Negative Integer**

**Output Format:**
**1 - If pair exists**

**0 - If no pair exists**

**Explanation for the given Sample Testcase:**

**YES as 5 - 1 = 4**

**So Return 1.**

**For example:**

| Input | Result |
|---|---|
| 3<br>1 3 5<br>4 | 1 |

**Answer:**

```
#include <stdio.h>


int main() {
   int n, k;


   scanf("%d", &n);


   int a[n];


   for (int i = 0; i < n; i++)
      scanf("%d", &a[i]);


   scanf("%d", &k);


   for (int i = 0; i < n; i++) {
      for (int j = 0; j < n; j++) {
         if (i != j && a[j] - a[i] == k) {
            printf("1");
            return 0;
         }
      }
   }


   printf("0");


   return 0;
}
```

**RESULT:**

Thus the code is verified and the expected output is obtained.